CELL STRUCTURE

Introduction

The cell is the fundamental unit of life, both structurally and functionally. All living organisms are composed of cells, and every physiological and biochemical activity essential for life occurs within them. The concept of the cell as the basic unit was established through the cell theory, which states that all organisms are made up of cells and that new cells arise from pre-existing cells. Understanding cell structure is essential for NEET, as many questions are directly based on organelle functions and differences between cell types.

Types of Cells

Cells are classified into prokaryotic and eukaryotic types based on internal organization. Prokaryotic cells lack a true nucleus and membrane-bound organelles, with their genetic material present in a nucleoid region. These cells are simple, smaller in size, and include bacteria and cyanobacteria.

Eukaryotic cells possess a well-defined nucleus enclosed within a nuclear membrane and contain specialized organelles. These cells are larger and more complex, forming the basis of plant and animal systems. NEET questions often test the structural differences and functional advantages of eukaryotic cells over prokaryotic cells.

Plasma Membrane

The plasma membrane forms the outer boundary of the cell and regulates the movement of substances. It follows the fluid mosaic model, consisting of a phospholipid bilayer with embedded proteins. This arrangement provides both flexibility and selective permeability.

Transport across the membrane occurs through passive processes such as diffusion and osmosis, as well as active transport that requires energy. Bulk transport mechanisms like endocytosis and exocytosis are also important. In NEET, questions frequently focus on membrane structure and transport mechanisms.

Cell Wall

The cell wall is present in plant cells and provides structural support and protection. It is primarily composed of cellulose. It maintains the shape of the cell and prevents excessive expansion when water enters, thereby protecting the cell from bursting. The absence of a cell wall in animal cells is an important distinguishing feature often asked in exams.

Cytoplasm

The cytoplasm is the semi-fluid matrix that contains all cell organelles. It is the site of various metabolic reactions and plays a crucial role in maintaining the internal organization of the cell. Movement of substances and coordination of cellular activities occur within this region.

Endoplasmic Reticulum

The endoplasmic reticulum is a network of membranes connected to the nuclear envelope. It exists in two forms: rough and smooth. Rough endoplasmic reticulum contains ribosomes and is involved in protein synthesis, whereas smooth endoplasmic reticulum is responsible for lipid synthesis and detoxification.

This organelle is important in intracellular transport, and NEET questions often focus on the functional differences between rough and smooth ER.

Golgi Apparatus

The Golgi apparatus consists of stacked, flattened sacs and plays a key role in modifying, packaging, and transporting proteins and lipids. It also forms lysosomes and participates in secretion. Its role in protein processing is a frequently tested concept in NEET.

Lysosomes

Lysosomes are membrane-bound organelles containing hydrolytic enzymes capable of breaking down complex substances. They are involved in intracellular digestion and removal of waste materials. Due to their ability to digest cellular components, they are often referred to as suicide bags, a concept commonly asked in exams.

Mitochondria

Mitochondria are double-membraned organelles responsible for energy production through cellular respiration. The inner membrane forms cristae, increasing the surface area for enzymatic reactions. The matrix contains enzymes necessary for ATP synthesis. They are known as the powerhouse of the cell and are frequently asked in NEET questions.

Ribosomes

Ribosomes are non-membrane-bound structures that serve as the sites of protein synthesis. They can be found free in the cytoplasm or attached to the rough endoplasmic reticulum. Their role in protein synthesis makes them essential for cell function.

Plastids

Plastids are present in plant cells and are involved in food synthesis and storage. Chloroplasts contain chlorophyll and carry out photosynthesis, while chromoplasts provide color and leucoplasts store food. Chloroplast structure and function are important topics for NEET.

Vacuole

The vacuole is a membrane-bound structure that stores cell sap. In plant cells, it is large and helps maintain turgor pressure, which is essential for maintaining rigidity. In animal cells, vacuoles are smaller or absent. Questions often test this structural difference.

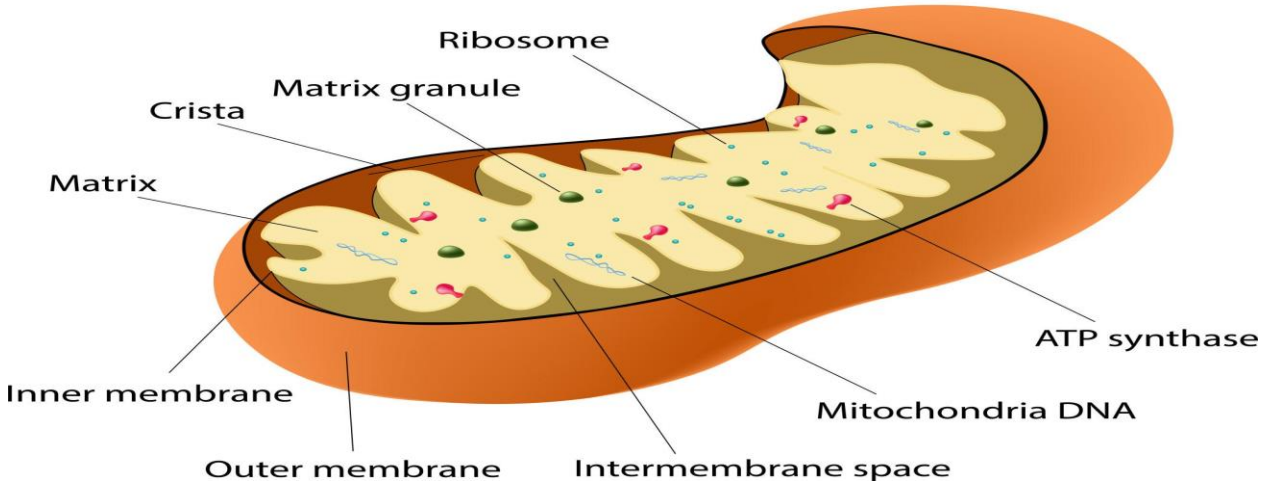
Nucleus

The nucleus is the control center of the cell and contains genetic material. It is enclosed by a nuclear membrane and contains nucleoplasm, chromatin, and nucleolus. Chromatin condenses into chromosomes during cell division. The nucleus regulates all cellular activities, including and reproduction.

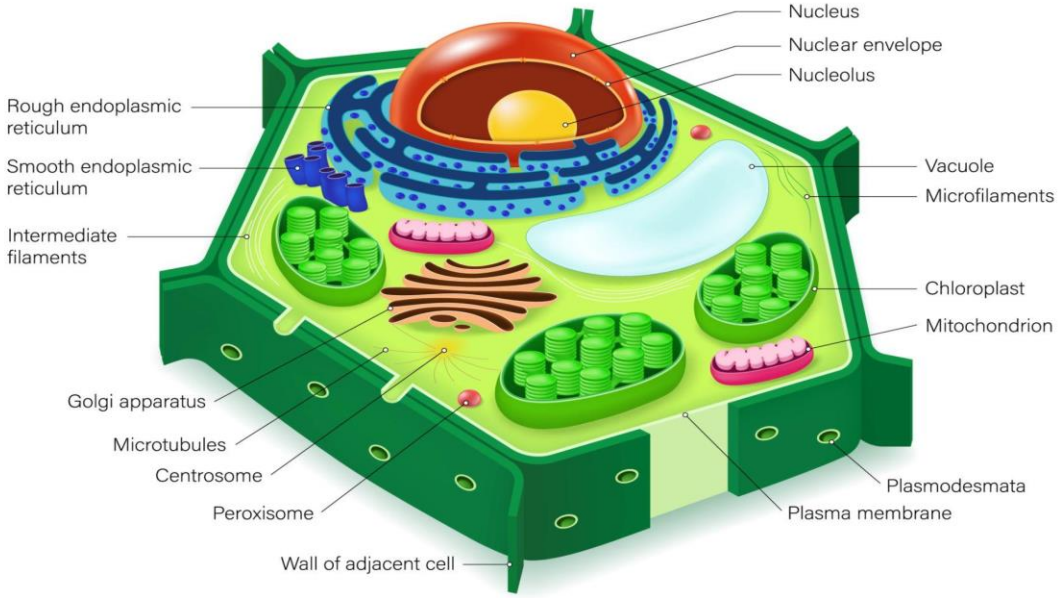
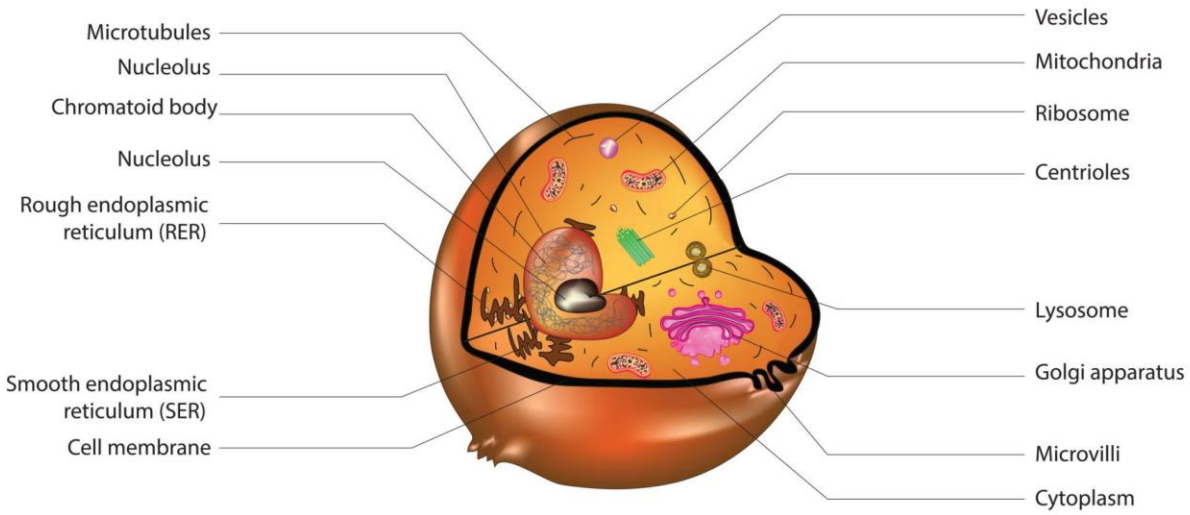
Plant Cell and Animal Cell

Plant cells are characterized by the presence of a cell wall, plastids, and a large vacuole, which provide rigidity and support. Animal cells lack these structures and have a more flexible shape. Understanding these differences is important for NEET.

Mitochondria



Plant cell



Structure of Animal Cell